EXP-2 Bias Audit on Word Embeddings using WEAT

AIM:

To perform a Bias Audit on pre-trained word embeddings to identify gender or racial stereotypes using the Word Embedding Association Test (WEAT).

DATASET:

Pre-trained Word Embeddings (Google News Word2Vec / Common Crawl GloVe)

PROCEDURE:

Step 1: Install and Import Libraries

```
!pip install gensim numpy scipy
import numpy as np
from scipy.spatial.distance import cosine
import gensim.downloader as api
```

Step 2: Load Pre-trained Word Embeddings

Load Google News Word2Vec embeddings (large model)

```
model = api.load("word2vec-google-news-300")
```

Step 3: Define Target and Attribute Word Sets

Target words

```
male_words = ["man", "boy", "father", "he", "son"] female_words = ["woman", "girl", "mother", "she", "daughter"]
```

Attribute words

```
career_words = ["career", "corporation", "salary", "office", "professional"] family_words = ["home", "family", "children", "kitchen", "marriage"]
```

Step 4: Define Cosine Similarity Function

```
def cosine_similarity(w1, w2):
    return 1 - cosine(model[w1], model[w2])
```

Step 5: Association Function (WEAT Core)

```
def association(w, A, B):
return np.mean([cosine_similarity(w, a) for a in A]) - \
np.mean([cosine_similarity(w, b) for b in B])
```

Step 6: Compute WEAT Test Statistic

```
def weat_statistic(X, Y, A, B):

return sum(association(x, A, B) for x in X) - sum(association(y, A, B) for y in Y)
weat_score = weat_statistic(male_words, female_words, career_words, family_words)
weat_score
```

Step 7: Effect Size Calculation

```
def effect_size(X, Y, A, B):

assoc_X = [association(x, A, B) for x in X]
assoc_Y = [association(y, A, B) for y in Y]
 mean_diff = np.mean(assoc_X) - np.mean(assoc_Y)
std_dev = np.std(assoc_X + assoc_Y)
return mean_diff / std_dev

 effect = effect_size(male_words, female_words, career_words, family_words)
effect
```

Step 8: Permutation Test (Statistical Significance)

```
import random

def permutation_test(X, Y, A, B, iterations=1000):

combined = X + Y

observed = weat_statistic(X, Y, A,
B) count = 0
for _ in range(iterations):

random.shuffle(combined) X_perm =
combined[:len(X)] Y_perm =
```

combined[len(X):]

perm_stat = weat_statistic(X_perm, Y_perm, A, B) if

abs(perm_stat) > abs(observed):

count += 1

return count / iterations

p_value = permutation_test(male_words, female_words, career_words, family_words) p_value

Output:

```
Requirement already satisfied: gensim in /usr/local/lib/python3.12/dist-packages (4.4.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.2)
Loading Google News Word2Vec (takes time)...
Model loaded successfully!

===== WEAT RESULTS =====
WEAT Score: 0.417
Effect Size: 0.801
P-value: 0.25

===== INTERPRETATION =====
Bias: Male → Career, Female → Family
Strong Bias Detected
Not Statistically Significant
```

RESULT:

Bias exists in pre-trained word embeddings, demonstrating that models trained on large text corpora inherit **human-like stereotypes**, which must be mitigated in AI systems.